

How to Build a Rhythm Game in Godot

GodotCon Boston 2026





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Who am I?

- Games and tools developer
- Worked with Unity, Unreal and Godot
- Built a lot of what I work on as open source projects
- Currently building my own game engine (in SDL)
- And a multi-engine tool for building rhythm games (Unity, Unreal, Godot, MonoGame, SDL, etc)
- A cat dad (most important).



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Outline of Workshop

1. Intro to Rhythm Game Utilities plugin
2. Intro to song charting
3. Workshop

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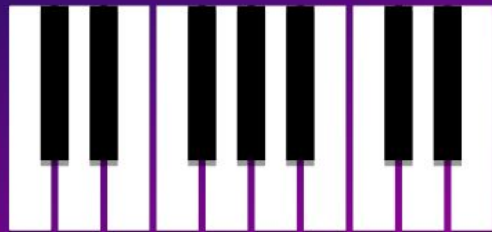
Intro to Rhythm Game Utilities plugin

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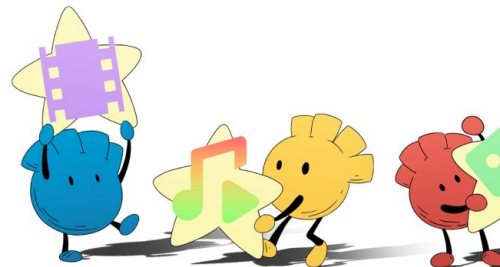
Rhythm Game Utilities

A collection of utilities for creating rhythm games in
Unity, Unreal, Godot, SDL and MonoGame.



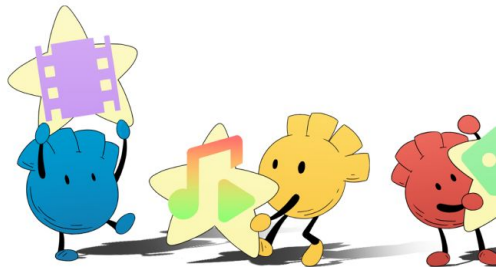
<https://rhythmgameutilities.com/>

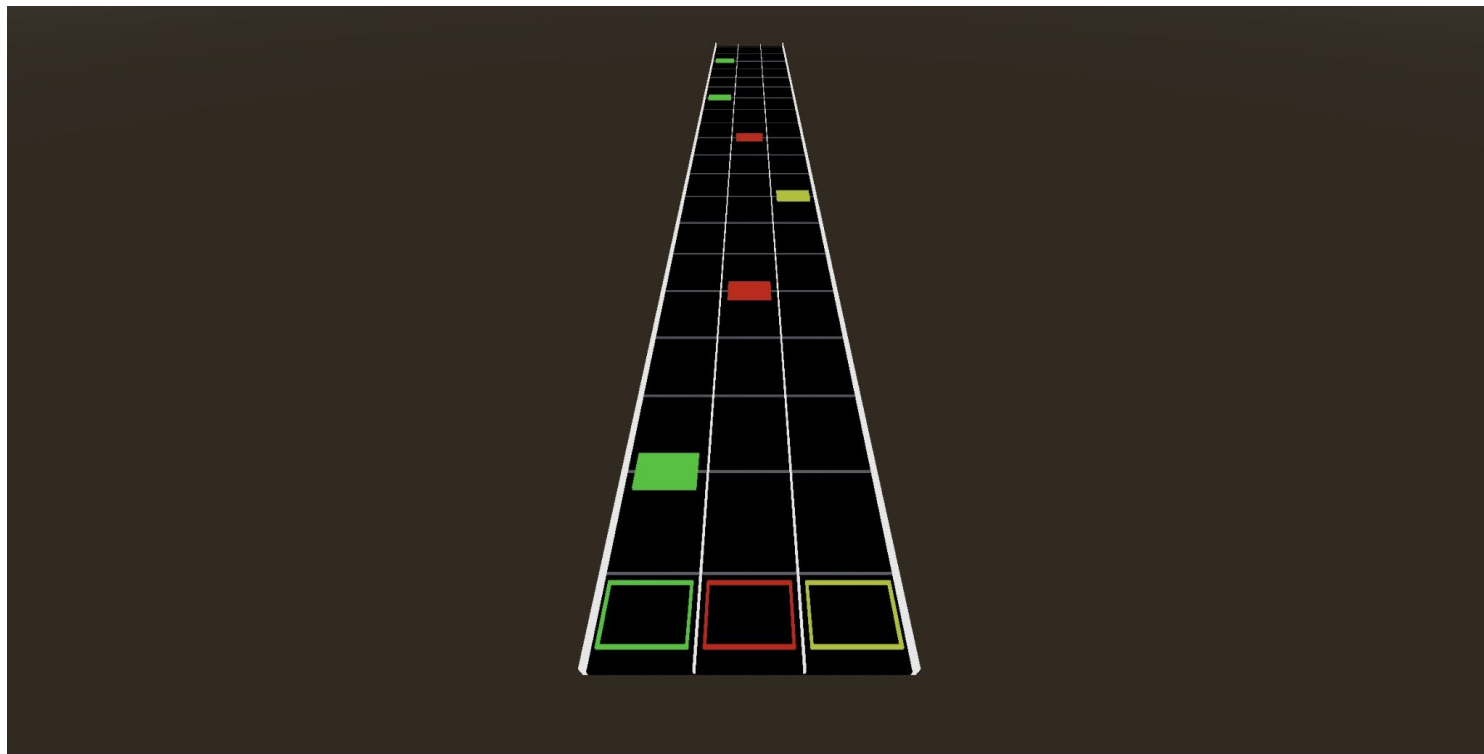
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Intro to Rhythm Game Utilities plugin

- Parse **.chart** and **.mid** files
- Calculate positions to render notes
- Calculate positions to render beat bars
- Find notes near the current time
- Calculate hit accuracy (*enum*)
- Calculate hit accuracy ratio (*float*)
- Determine if the current time is on the beat
- Generate data for rendering a waveform for an audio file





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Intro to song charting

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Intro to rhythm game data

Charting Songs

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Charting songs



notes.chart

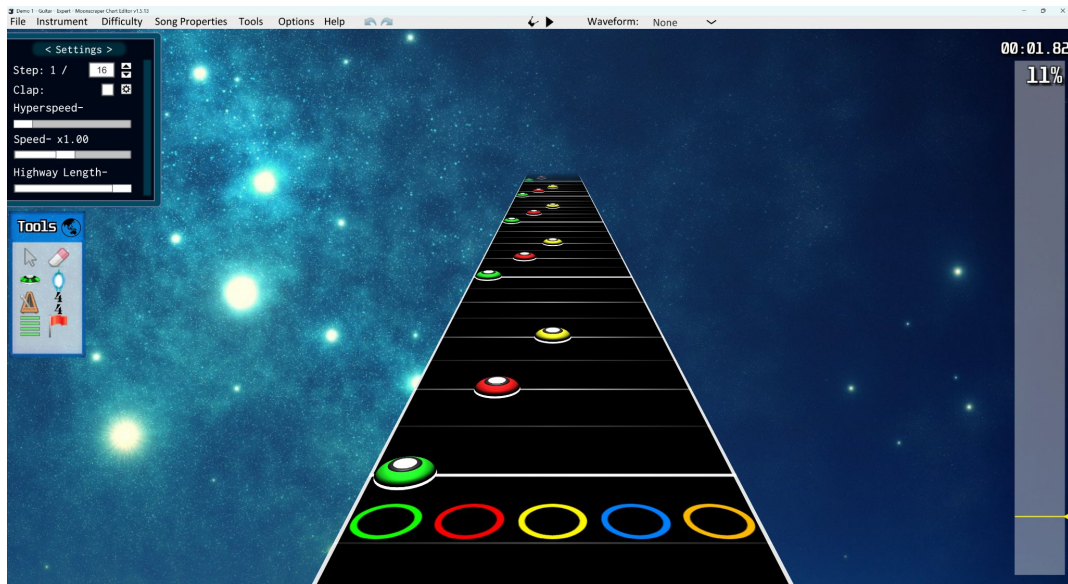


notes.mid

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Charting songs (.chart files)

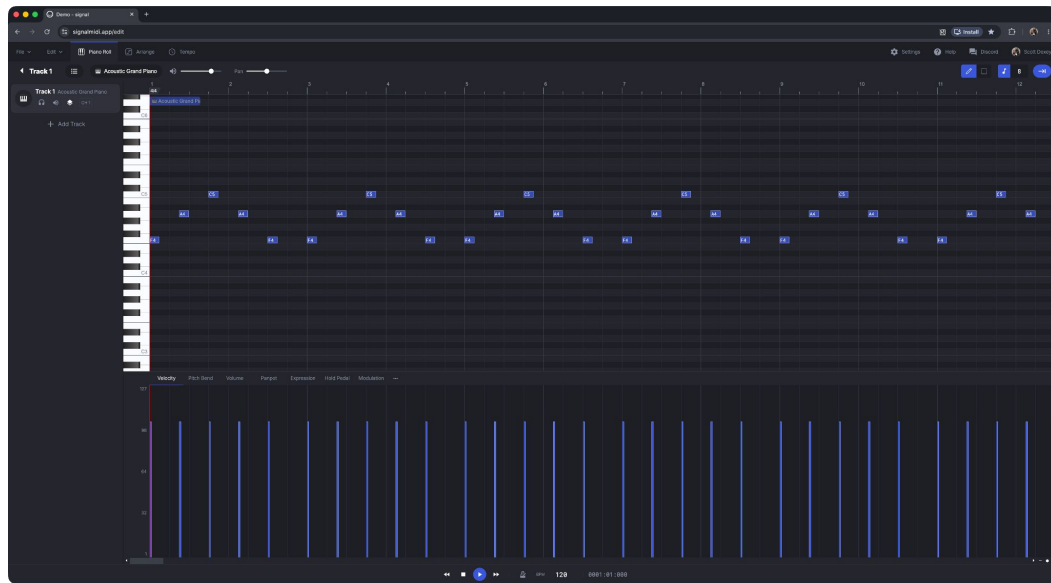


<https://github.com/FireFox2000000/Moonscraper-Chart-Editor>

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Charting songs (.mid files)



<https://signalmidi.app/>

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Intro to rhythm game data

Loading Songs

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Loading Notes (.chart)



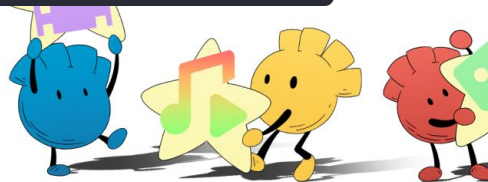
notes.chart



```
[Song]
{
  Name = "Demo 1"
  Artist = "Scott Doxey"
  Charter = "Scott Doxey"
  Album = "Demo"
  Year = ", 2024"
  Offset = 0
  Resolution = 192
  Player2 = bass
  Difficulty = 0
  PreviewStart = 0
  PreviewEnd = 0
  Genre = "rock"
  MediaType = "cd"
  MusicStream = "song.ogg"
}

[SyncTrack]
{
  0 = TS 4
  0 = B 120000
}

[ExpertSingle]
{
  768 = N 0 0
  960 = N 1 0
  1152 = N 2 0
  1536 = N 0 0
  1728 = N 1 0
  1920 = N 2 0
  2304 = N 0 0
  2496 = N 1 0
  2688 = N 2 0
  3072 = N 0 0
  3264 = N 1 0
  3456 = N 2 0
  3840 = N 0 0
  4032 = N 1 0
  4224 = N 2 0
  4608 = N 0 0
  4800 = N 1 0
  4992 = N 2 0
  5376 = N 0 0
  5568 = N 1 0
  5760 = N 2 0
}
```



Loading Notes (.chart)

```
[SyncTrack]
```

```
{
```

```
0 = TS 4
```

```
0 = B 120000
```

```
}
```

```
[ExpertSingle]
```

```
{
```

```
768 = N 0 0
```

```
960 = N 1 0
```

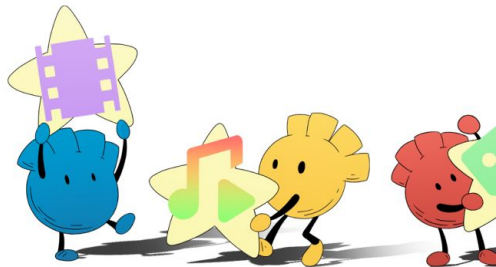
```
1152 = N 2 0
```

```
1536 = N 0 0
```

```
...
```

```
}
```

First value is the
tick (position)



Loading Notes (.chart)

```
[SyncTrack]
{
  0 = TS 4           TS = Time Signature
  0 = B 120000       B = BPM
}
```



Loading Notes (.chart)

```
[ExpertSingle]
```

```
{
```

```
768 = N 0 0
```

```
960 = N 1 0
```

```
1152 = N 2 0
```

```
1536 = N 0 0
```

```
...
```

```
}
```

Note
Hand Position
Length (Hold Notes)

N 1 0



Loading Notes



notes.chart



notes.mid

```
{
  "Name": "Demo 1",
  "Artist": "Scott Doxey",
  "Album": "Demo",
  "Genre": "rock",
  "Year": " ", 2024,
  "Charter": "Scott Doxey",
  "Resolution": 192,
  "Difficulty": 0,
  "Offset": 0,
  "PreviewStart": 0,
  "PreviewEnd": 0,
  "MusicStream": "song.ogg",
  "Lyrics": {},
  ...
}

{
  ...,
  "BPM": {
    "0": 120000
  },
  "TimeSignatures": {
    "0": [
      4
    ]
  }
}

{
  ...,
  "Difficulties": {
    ...,
    "Expert": [
      {
        "Position": 768,
        "HandPosition": 0,
        "Length": 0
      },
      {
        "Position": 960,
        "HandPosition": 1,
        "Length": 0
      },
      {
        "Position": 1152,
        "HandPosition": 2,
        "Length": 0
      },
      {
        "Position": 1536,
        "HandPosition": 0,
        "Length": 0
      },
      ...
    ],
    ...
  },
  ...
}
```



Disclaimer:

The demo we are building is not optimized for performance or using best practices.

The code is optimized for this workshop to make understanding the concepts easier in the restricted time frame.

This demo is without juice.



Workshop

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Questions?

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